For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Montemac 5

Montelukast Chewable Tablets 5 mg

COMPOSITION

Each chewable tablet contains:
Montelukast Sodium USP 5.2 mg
Equivalent to Montelukast…...5 mg

DESCRIPTION

Montelukast sodium is leukotriene receptor antagonist.
Chemically, Montelukast sodium is [R-(E)]-1-[[[1-[3-[2-(7-chloro-2quinolinyl)ethenyl]phenyl]-3-[2-(1-hydroxy-1 -methylethyl)phenyl]propyl]thio]methyl] cyclopropane acetic acid, monosodium salt. Its molecular formula is $C_{35}H_{35}ClNNaO_3S$, and molecular weight is 608.18.

PHARMACEUTICAL FORM

Pink color, mottled, circular, biconvex, uncoated chewable tablet debossed with 'CL 56' on one side and plain on other side.

PHARMACOLOGICAL CLASSIFICATION

Leukotriene Receptor Antagonist

PHARMACOLOGICAL ACTION:

Pharmacokinetics

Absorption : Montelukast is rapidly absorbed following oral administration. The mean oral bioavailability is 64%. The oral bioavailability and Cmax are not influenced by a standard meal. For the 5 mg chewable tablet, the Cmax is achieved in two hours after administration in adults in the fasted state. The mean oral bioavailability is 73% and is decreased to 63% by a standard meal. After administration of the 4 mg chewable tablet to paediatric patients 2 to 5 years of age in the fasted state, Cmax is achieved 2 hours after administration. The mean Cmax is 66% higher while mean Cmin is lower than in adults receiving a 10 mg tablet.

Distribution : Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 litres. Biotransformation Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children. Cytochrome P450 2C8 is the major enzyme in the metabolism of montelukast. Additionally CYP 3A4, and 2C9 may have a minor contribution, although itraconazole, an inhibitor of CYP 3A4, was shown not to change pharmacokinetic variables of montelukast. Based on in vitro results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

Elimination : The plasma clearance of montelukast averages 45 ml/min in adults. Following an oral dose of radiolabelled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

Characteristics in Patients : No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Studies in patients with renal impairment have not been undertaken. Because montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. There are no data on the pharmacokinetics of montelukast in patients with severe hepatic insufficiency (Child-Pugh score >9). With high doses of montelukast (20- and 60-fold the recommended adult dose), a decrease in plasma theophylline concentration was observed. This effect was not seen at the recommended dose of 10 mg once daily.

Pharmacodynamics

Pharmacotherapeutic group: Leukotriene Receptor Antagonist
ATC Code:R03D C03
Mode of action:

The cysteinyl leukotrienes (LTC4, LTD4, LTE4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils.
These important proasthmatic mediators bind to cysteinyl leukotriene receptors (CysLT) found in the human airway and cause airway actions, including bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. Montelukast is an orally active compound which binds with
high affinity and selectivity to the CysLT1 receptor. In clinical studies, montelukast inhibits bronchoconstriction due to inhaled LTD4 at doses as low as 5 mg. Bronchodilation was observed within two hours of oral administration. The bronchodilation effect caused by a βagonist was additive to that caused by montelukast. Treatment with montelukast inhibited both early and late phase bronchoconstriction due to antigen challenge. Montelukast, compared with placebo, decreased peripheral blood eosinophils in adult and paediatric patients. In a separate study, treatment with montelukast significantly decreased eosinophils in the airways (as measured in sputum) and in peripheral blood while improving clinical asthma control.

INDICATIONS AND USAGE

Montelukast is indicated for the prophylaxis and chronic treatment of asthma in adults and pediatric patients 12 months of age and older. Montelukast is indicated in adults and pediatric patients 2 years of age and older for the relief of daytime and nighttime symptoms of seasonal allergic rhinitis.

CONTRAINDICATIONS

Hypersensitivity to any component of this product.

ADVERSE REACTION

System Organ Class	Adverse experience term
Infections and infestations	Upper respiratory infection
Blood any lymphatic system disorders	Increased bleeding tendency
Immune system disorders	Hypersensitivity reactions including anaphylaxis
	hepatic eosinophilic infiltration
Psychiatric disorders	dream abnormalities including nightmares, insomnia, somnambulism, irritability, anxiety, restlessness, agitation including aggressive behaviour or hostility, depression
	Disturbance in attention, memory impairment
	hallucinations, disorientation, suicidal thinking and behaviour (suicidality)
Nervous system disorders	Diziness, drowsiness, Paraesthesia/hypoesthesia, seizure
Cardiac disorders	Palpitations
Respiratory, thoracic and mediastinal disorders	Epistaxis Churg-Strauss Syndrome (CSS)
Gastrointestinal disorders	Diarrhoea, nausea, vomiting
	dry mouth, dyspepsia
Hepatobiliary disorders	elevated levels of serum transaminases (ALT, AST)
	hepatitis (including cholestatic, hepatocellular, and mixed pattern liver injury)
Skin and subcutaneous tissue disorders	Rash
	bruising, urticaria, pruritus
	angiooedema
	erythema nodosum, erythema multiforme
Muskosceletal and connective tissue disorders	Arthralgia, myalgia, including muscular cramps
General disorders and administration site conditions	Pyrexia
	Asthenia/fatigue, malaise, oedema

Postmarketing Experience :

Blood and lymphatic system disorders :
thrombocytopenia

DOSAGE

MONTEMAC should be taken once daily.
For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Adults 15 Years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis: The dosage for adults 15 years of age and older is one 10-mg tablet daily.

Pediatric Patients 2 to 5 Years of Age with Asthma and/or Seasonal Allergic Rhinitis: The dosage for pediatric patients 2 to 5 years of age is one 4-mg chewable tablet daily.

General Recommendations: The therapeutic effect of MONTEMAC on parameters of asthma control occurs within one day. MONTEMAC tablets, chewable tablets, and oral granules can be taken with or without food. Patients should be advised to continue taking MONTEMAC while their asthma is controlled, as well as during periods of worsening asthma. No dosage adjustment is necessary for pediatric patients, for the elderly, for patients with renal insufficiency, or mild-to-moderate
hepatic impairment, or for patients of either gender.
Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent. Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.
Therapy with MONTEMAC in Relation to Other Treatments for Asthma: MONTEMAC can be added to a patient's existing treatment regimen.
Reduction in Concomitant Therapy: Bronchodilator Treatments: MONTEMAC can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.
Inhaled Corticosteroids: Treatment with MONTEMAC provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. MONTEMAC should not be abruptly substituted for inhaled corticosteroids.

DRUG INTERACTIONS

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following medicinal products: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl oestradiol/norethindrone 35/1), terfenadine, digoxin and warfarin.
The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. Since montelukast is metabolised by CYP 3A4, 2C8, and 2C9, caution should be exercised, particularly in children, when montelukast is co-
administered with inducers of CYP 3A4, 2C8, and 2C9, such as phenytoin, phenobarbital and rifampicin.
In vitro studies have shown that montelukast is a potent inhibitor of CYP 2C8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of medicinal products primarily metabolised by CYP 2C8) demonstrated that montelukast does not inhibit CYP 2C8 *in vivo.* Therefore, montelukast is not anticipated to markedly alter the metabolism of medicinal products metabolised by this enzyme (eg.,paclitaxel, rosiglitazone, and repaglinide).
In vitro studies have shown that montelukast is a substrate of CYP 2C8, and to a less significant extent, of 2C9, and 3A4. In a clinical drug-drug interaction study involving montelukast and gemfibrozil (an inhibitor of both CYP 2C8 and 2C9) gemfibrozil increased the systemic exposure of montelukast by 4.4-fold. No routine dosage adjustment of montelukast is required upon co-administration with gemfibrozil or other potent inhibitors of CYP 2C8, but the physician should be aware of the potential for an increase in adverse reactions.
Based on *in vitro* data, clinically important drug interactions with less potent inhibitors of CYP 2C8 (e.g.,trimethoprim) are not anticipated. Co administration of montelukast with itraconazole, a strong inhibitor of CYP 3A4, resulted in no significant increase in the systemic exposure of montelukast.

WARNINGS & PRECAUTIONS

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β-agonist should be used. Patients should seek their doctor's advice as soon as possible if they need more inhalations of short-acting β-agonists than usual. Montelukast should not be abruptly substituted for inhaled or oral corticosteroids. There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly. In rare cases, patients on therapy with antiasthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.
Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other nonsteroidal antiinflammatory drugs. Unsuitable for phenylketonurics.

Effects on ability to drive and use machines

Montelukast is not expected to affect a patient's ability to drive a car or operate machinery. However, in very rare cases, individuals have reported drowsiness or dizziness.

Pregnancy and Lactation

Use during pregnancy

Animal studies do not indicate harmful effects with respect to effects on pregnancy or embryonal/foetal development. Limited data from available pregnancy databases do not suggest a causal relationship between Montemac and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience. Montemac may be used during pregnancy only if it is considered to be clearly essential.

Use during lactation

Studies in rats have shown that montelukast is excreted in milk (see section 5.3). It is not known if montelukast is excreted in human milk. Montemac may be used in breastfeeding mothers only if it is considered to be clearly essential.

SYMPTOMS AND TREATMENT OF OVERDOSE

No specific information is available on the treatment of overdose with montelukast. The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

It is not known whether montelukast is dialysable by peritoneal or haemodialysis.

STORAGE

Store below 30°C in a dry place, Protect from moisture & light.

KEEP OUT OF REACH OF CHILDREN.

PRESENTATION

Alu/Alu Blister pack of 10 Tablets. Such 10 blisters are packed in a carton along with a pack insert.

For further information please consult your physician or pharmacist.

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